Arpit Raj

Bachelor of Technology
Electronics & Communications Engineering
Shri Mata Vaishno Devi University
Katra, Jammu and Kashmir – 182320

184 Sector 10, Gurugram, India – 122001
+91 8700725192
rajrpit812@gmail.com
ArpitRaj
rajarpit812

Education

Degree/Certificate	Institute/Board	Year
B.Sc Data Science and Applications	Indian Institute of Technology	2025–Present
B.Tech ECE	Shri Mata Vaishno Devi University	2024–Present

Experience & Leadership

Team Lead, PCB Design Workshop 2025, *SMVDU*
Led a team in designing and fabricating a Printed Circuit Board (PCB) using Eagle, overseeing schematic design, layout, and manufacturing process.

Research Intern – Winter Consulting Project 2025, *IIT Guwahati*
Conducted data-driven analysis on OTT viewer retention using Python and exploratory data analysis techniques; identified behavioral patterns and proposed actionable strategies to improve engagement and reduce churn.

Technical Volunteer – Live Streaming 2025, *SMVDU Fest*
Managed live event streaming using OBS, handling multi-camera feeds and ensuring stable broadcast quality during large-scale university events.

Projects

AI Chatbot Platform 2026
Tech: React, Node.js, Express.js, MongoDB, Socket.io, Gemini API
Built a full-stack conversational AI platform with real-time messaging using Socket.io; implemented JWT-based authentication and integrated LLM APIs (Gemini, HuggingFace) for dynamic responses.

Driver Drowsiness Detection System 2026
Tech: Python, OpenCV, MediaPipe
Developed a real-time driver safety system using facial landmark detection and Eye Aspect Ratio (EAR) to detect drowsiness and trigger alerts.

Real-Time Hardware Image Processor 2025
Tech: Verilog, PYNQ-Z2 FPGA
Designed a high-speed video processing pipeline on FPGA for real-time image filtering and edge detection using custom hardware cores.

Virtual Air Drawing 2026
Tech: Python, OpenCV, MediaPipe
Built a touchless gesture-based drawing system using hand tracking, achieving 25+ FPS; designed for accessibility and interactive learning applications.

8-bit ALU Design 2025
Tech: Verilog
Designed and implemented an 8-bit ALU supporting multiple arithmetic and logical operations; verified functionality through simulation.

BJT-Based Wireless AC Current Detector 2024, *Hardware Design*
Designed and built a circuit using Bipolar Junction Transistors (BJT) to wirelessly detect the presence of AC current sources.

OTT Viewer Retention Analysis 2025, *IIT Guwahati Consulting Project*
Analyzed viewer engagement patterns in OTT platforms; identified key retention factors and proposed data-driven strategies to improve user engagement.

Technical Skills

Programming Languages	C, Python, SQL, Verilog, MATLAB
Web & Backend	HTML, CSS, Node.js, Express.js, MongoDB, Socket.io
Hardware & Embedded	PCB Design (Eagle), Arduino, Raspberry Pi, ESP-32, PYNQ-Z2 FPGA
Tools & Technologies	OpenCV, MediaPipe, Docker, Nginx, AWS, Vivado, Multisim
Other Tools	AutoCAD, Fusion 360, OBS Studio, Google Workspace, Microsoft Office

Soft Skills & Extra-curricular

Leadership	PR management in Aakash house of SMVDU, Lead C programming head, Lead Team of Printed Circuit Board Competition, Udyamitsav 2026 winner Finxcelerate.
Volunteering	NSS at SMVDU, Vikalp (A student initiative to educate unpriviledge kids), Ekatva Fest team 2025 and 2026, Udyamitsav 2026 IIT Jammu
Skills	Communication & Collaboration, Event Management, book fair

Leadership & Volunteering

Technical	Team Leader for PCB Design Workshop; Project Leader for C Programming applications.
Events	Core Member, Startup and Innovation Club (SMVDU); PR Manager & Organizer for Ekatva Fest (2025–2026); Core Member, Tarang Electronics Society; University Representative at JKEDI innovation program; Lead OBS live-streamer for university events.

2024–Present **Volunteer Teacher**, *Vikalp Initiative*, SMVDU
Provide free, consistent after-school tutoring in foundational subjects to underprivileged children from neighboring villages. Focus on bridging educational gaps, developing interactive lesson plans, and serving as a long-term mentor to foster a positive learning environment.

2024–Present **Core Contributor**, *Annapurna Initiative*, SMVDU
Organize and execute the systematic collection of surplus, hygienic food from university mess facilities. Manage the direct distribution of these meals to local daily-wage labor workers, actively working to minimize campus food waste while supporting vulnerable local populations.

2024–Present **Active Volunteer**, *National Service Scheme (NSS)*, SMVDU
Participate in university-led community outreach programs, health and hygiene awareness campaigns, and campus betterment drives. Collaborate with peers to organize events that promote social responsibility, civic engagement, and community welfare.

Certifications

Deloitte Australia - Technology Job Simulation 2025
textitForage

VLSI Design Course 2025
Internshala

J.P. Morgan - Software Engineering Job Simulation 2025
Forage

MATLAB Onramp 2025-26
MathWorks

Simulink Onramp 2025-26
MathWork

Consulting 2026
IIT Guwahati